

Scalar Kalman Filter

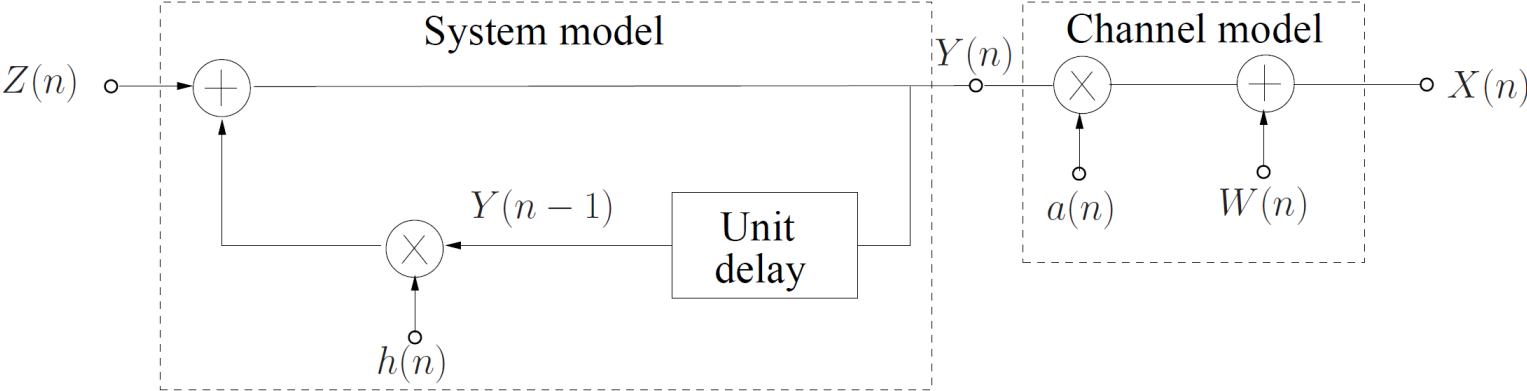
The Kalman Filter

Goal: to find LMMSE estimates of the samples of an unobservable process $Y(n)$ from the samples of a related, observable process $X(n)$, for $n = 1, 2, \dots$

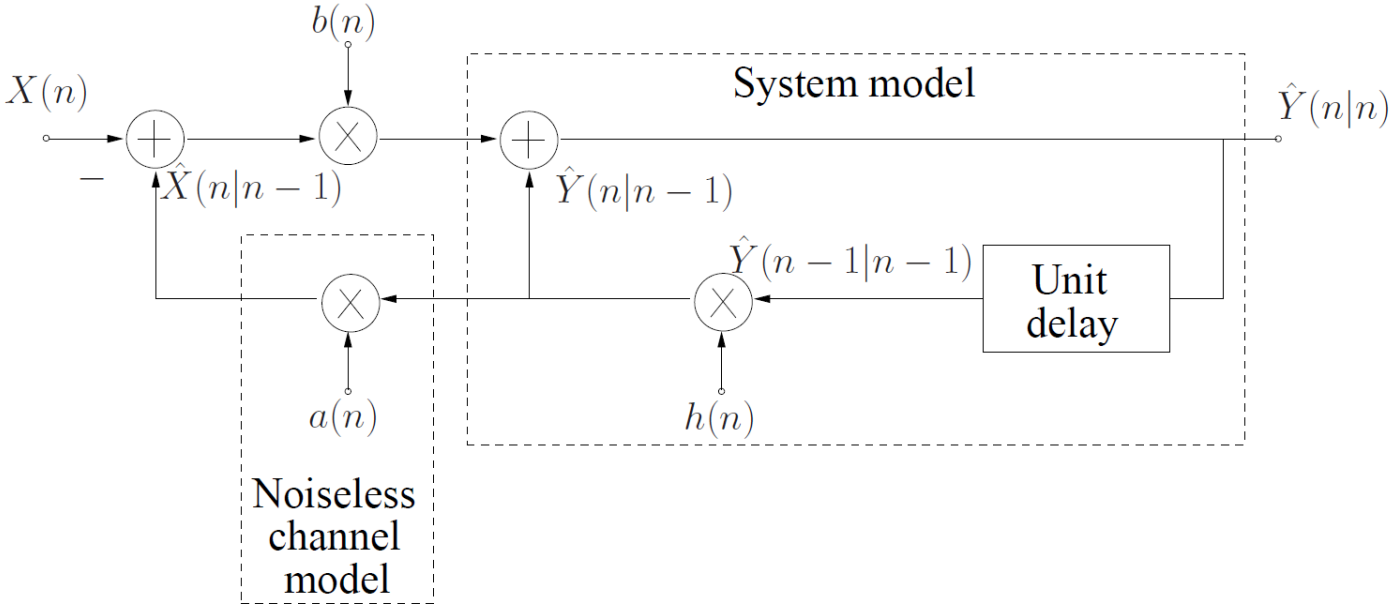
Recursive implementation: at time $n + 1$, the LMMSE estimate of $Y(n + 1)$ is obtained as a linear function of the LMMSE estimate of $Y(n)$ and the new observation $X(n + 1)$.

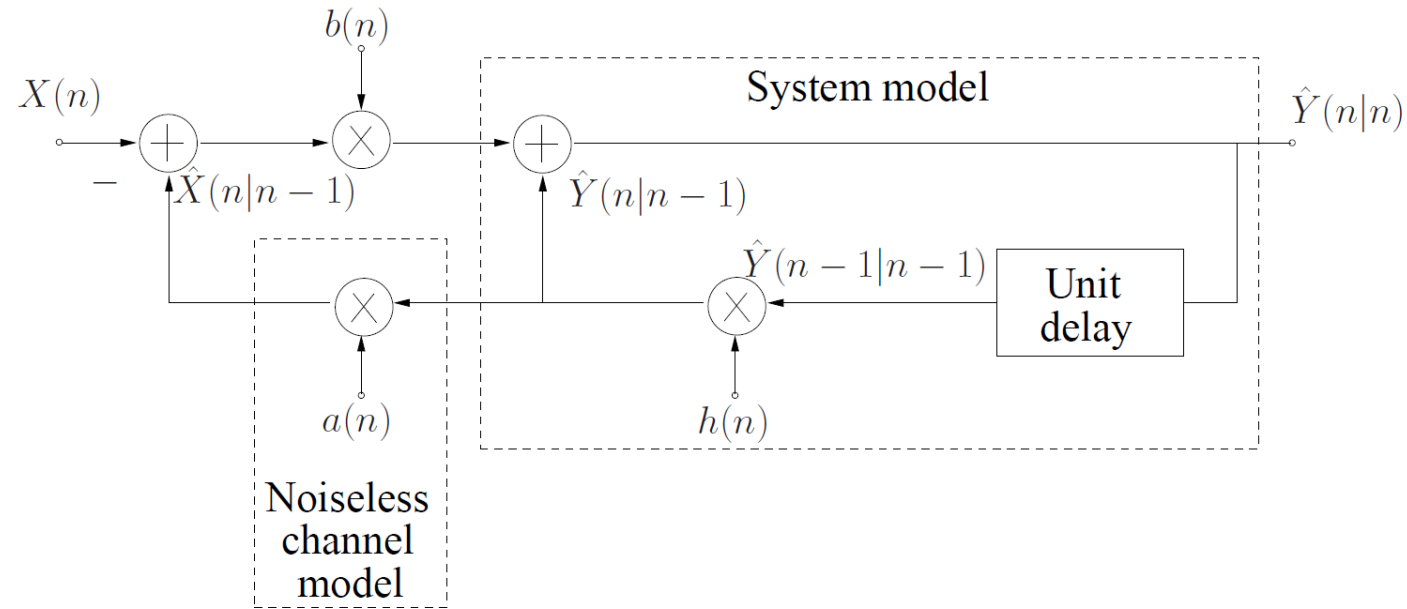
This recursive implementation is only possible when the processes $Y(n)$ and $X(n)$ are defined appropriately.

System and Channel Model
for the scalar Kalman Filter:



Block Diagram of the scalar
Kalman Filter:





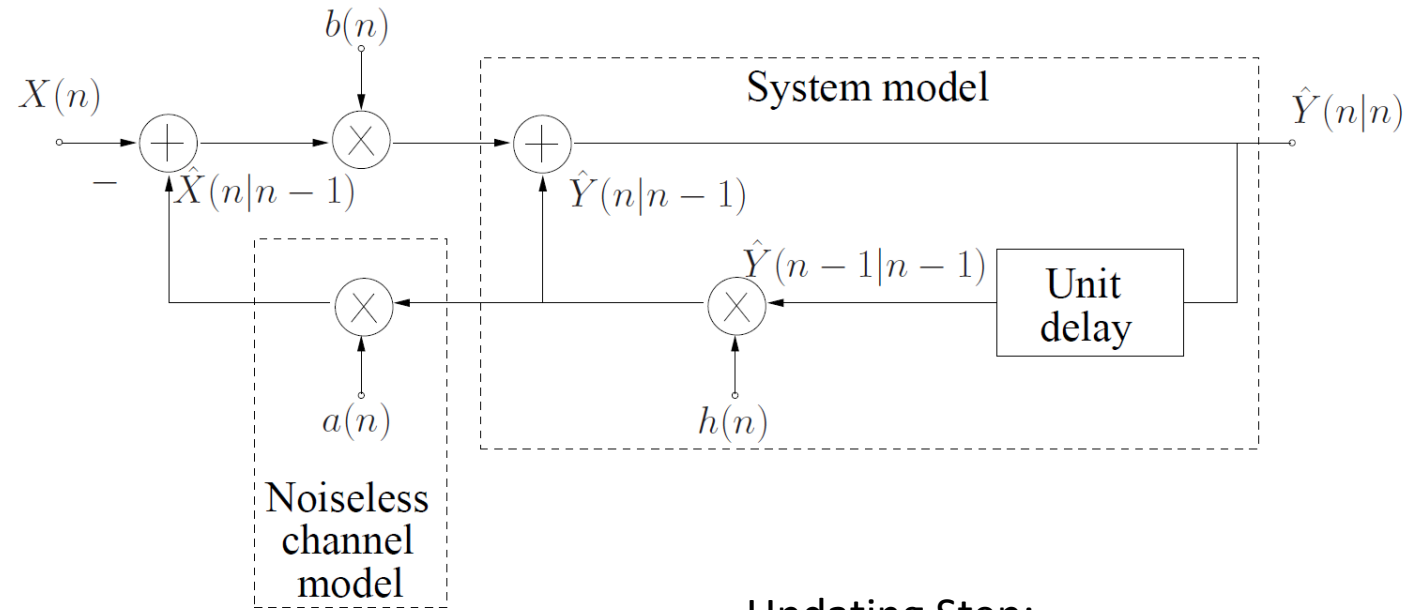
$\hat{Y}(n-1|n-1)$: LMMSE Estimate of $Y(n-1)$ given the observations $X(1), X(2), \dots, X(n-1)$.

$\hat{Y}(n|n-1)$: LMMSE Estimate of $Y(n)$ given the observations $X(1), X(2), \dots, X(n-1)$.

$\hat{X}(n|n-1)$: LMMSE Estimate of $X(n)$ given the observations $X(1), X(2), \dots, X(n-1)$.

$\hat{Y}(n|n)$: LMMSE Estimate of $Y(n)$ given the observations $X(1), X(2), \dots, X(n)$.

Recursive structure of the Kalman filter: $\hat{Y}(n|n) = \mathcal{LF}(\hat{Y}(n-1|n-1), X(n))$



Prediction Step:

$$\hat{Y}(n|n-1) = h(n)\hat{Y}(n-1|n-1),$$

$$\hat{X}(n|n-1) = a(n)\hat{Y}(n|n-1).$$

In addition, the MSE of $\hat{Y}(n|n-1)$ is:

$$R(n|n-1) = h(n)^2 R(n-1|n-1) + \sigma_Z^2(n)$$

Updating Step:

$$\hat{Y}(n|n) = \hat{Y}(n|n-1) + b(n)(X(n) - \hat{X}(n|n-1))$$

with the Kalman gain:

$$b(n) = \frac{a(n)R(n|n-1)}{a(n)^2 R(n|n-1) + \sigma_W^2(n)}.$$

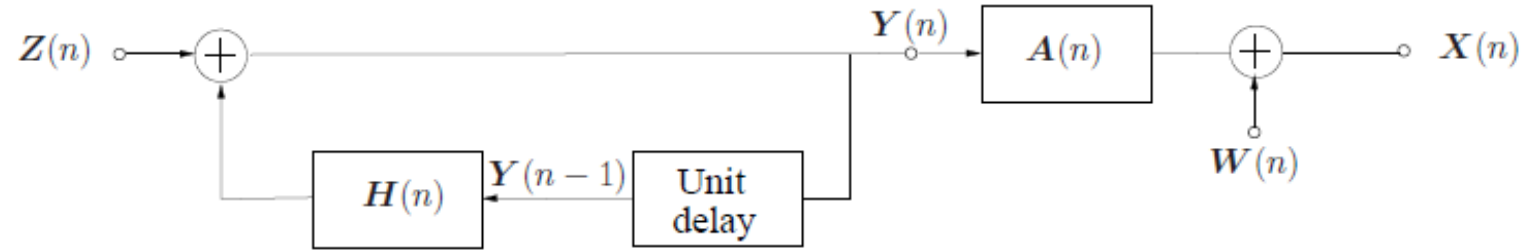
In addition, the MSE of $\hat{Y}(n|n)$ is:

$$R(n|n) = (1 - b(n)a(n))R(n|n-1)$$

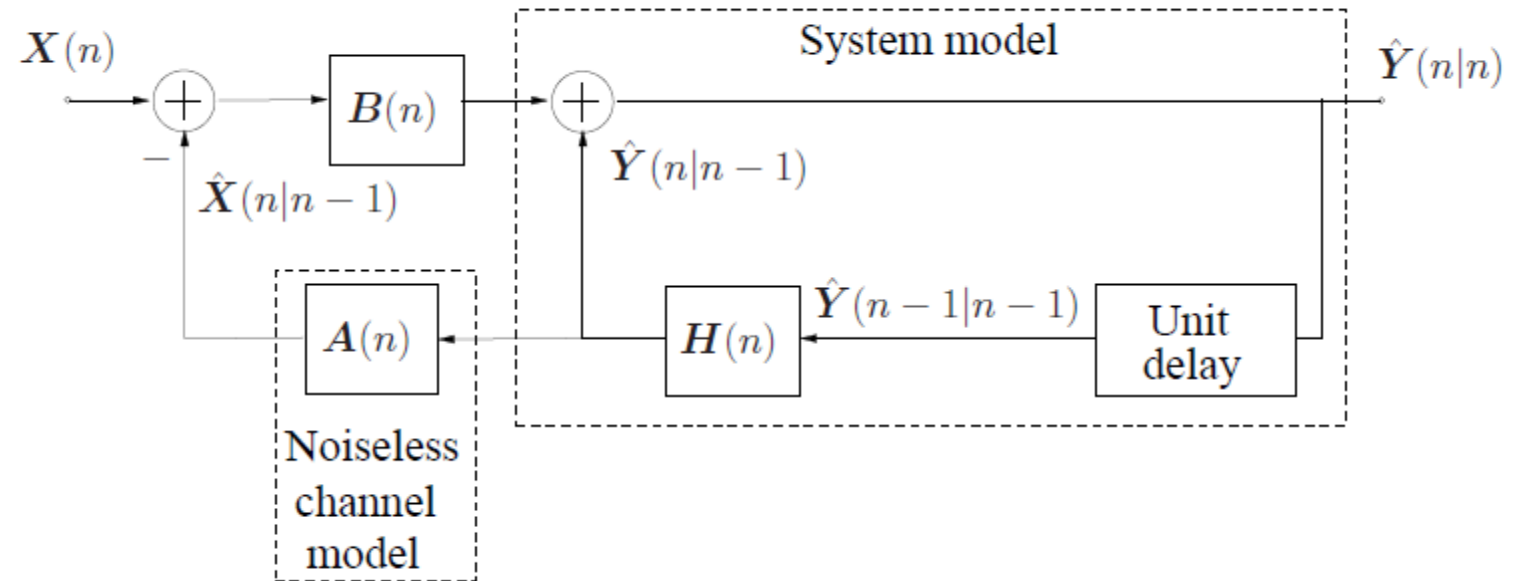
Initialization: the Kalman filter is initialized by setting $\hat{Y}(0|0) = \mu_{Y(0)}$ and $R(0|0) = \sigma_{Y(0)}^2$

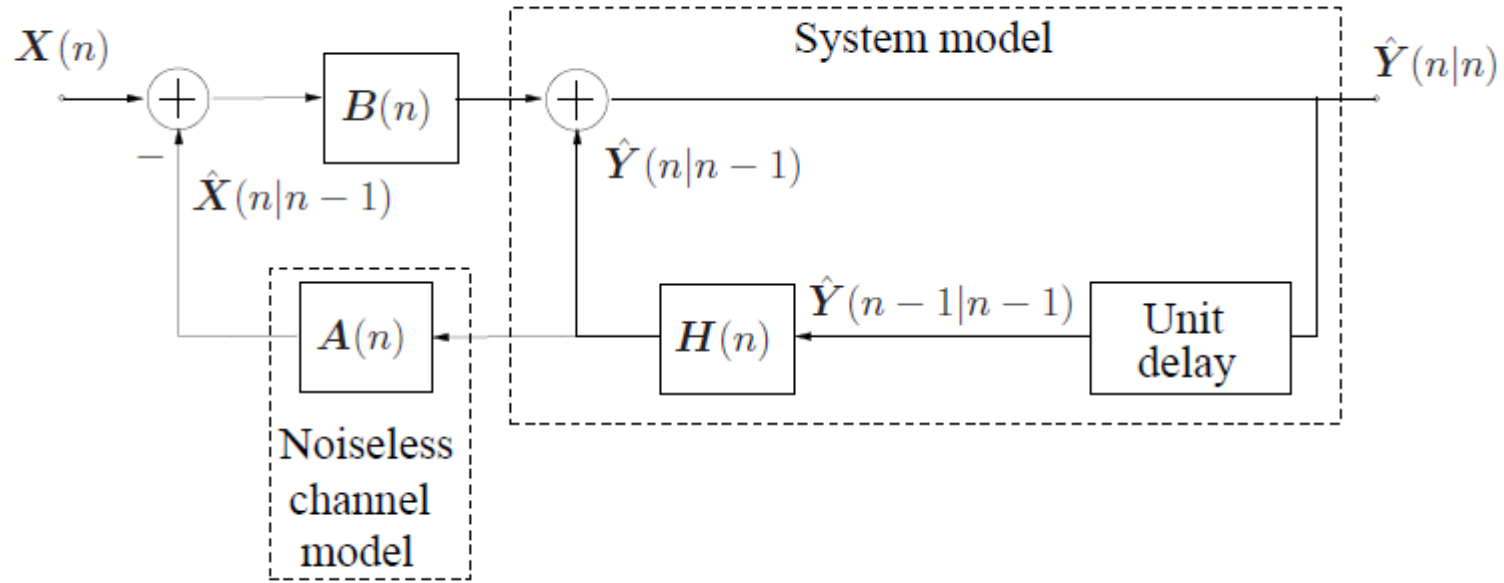
Vector Kalman Filter

System and Channel Model
for the vector Kalman
Filter:



Block Diagram of the
vector Kalman Filter:





Prediction Step:

$$\hat{\mathbf{Y}}(n|n-1) = \mathbf{H}(n)\hat{\mathbf{Y}}(n-1|n-1),$$

$$\hat{\mathbf{X}}(n|n-1) = \mathbf{A}(n)\hat{\mathbf{Y}}(n|n-1).$$

In addition, the covariance matrix of the estimation error of $\hat{\mathbf{Y}}(n|n-1)$ is:

$$\mathbf{R}(n|n-1) = \mathbf{H}(n)\mathbf{R}(n-1|n-1)\mathbf{H}(n)^T + \mathbf{Q}_Z(n)$$

Updating Step:

$$\hat{\mathbf{Y}}(n|n) = \hat{\mathbf{Y}}(n|n-1) + \mathbf{B}(n)(\mathbf{X}(n) - \hat{\mathbf{X}}(n|n-1))$$

with the Kalman gain:

$$\mathbf{B}(n) = \mathbf{R}(n|n-1)\mathbf{A}(n)^T[\mathbf{A}(n)\mathbf{R}(n|n-1)\mathbf{A}(n)^T + \mathbf{Q}_W(n)]^{-1}$$

In addition, the covariance matrix of the estimation error of $\hat{\mathbf{Y}}(n|n)$ is:

$$\mathbf{R}(n|n) = (\mathbf{I} - \mathbf{B}(n)\mathbf{A}(n))\mathbf{R}(n|n-1)$$

Initialization: the vector Kalman filter is initialized by setting $\hat{\mathbf{Y}}(0|0) = \boldsymbol{\mu}_{\mathbf{Y}(0)}$ and $\mathbf{R}(0|0) = \boldsymbol{\Sigma}_{\mathbf{Y}(0)\mathbf{Y}(0)}$

Comparison: Scalar / Vector
Kalman Filter

Comparison

Scalar Version

Prediction Step:

$$\hat{Y}(n|n-1) = h(n)\hat{Y}(n-1|n-1),$$

$$\hat{X}(n|n-1) = a(n)\hat{Y}(n|n-1).$$

$$R(n|n-1) = h(n)^2 R(n-1|n-1) + \sigma_Z^2(n)$$

Updating Step:

$$\hat{Y}(n|n) = \hat{Y}(n|n-1) + b(n)(X(n) - \hat{X}(n|n-1))$$

$$b(n) = \frac{a(n)R(n|n-1)}{a(n)^2 R(n|n-1) + \sigma_W^2(n)}.$$

$$R(n|n) = (1 - b(n)a(n))R(n|n-1)$$

Vector Version

Prediction Step:

$$\hat{\mathbf{Y}}(n|n-1) = \mathbf{H}(n)\hat{\mathbf{Y}}(n-1|n-1),$$

$$\hat{\mathbf{X}}(n|n-1) = \mathbf{A}(n)\hat{\mathbf{Y}}(n|n-1).$$

$$\mathbf{R}(n|n-1) = \mathbf{H}(n)\mathbf{R}(n-1|n-1)\mathbf{H}(n)^T + \mathbf{Q}_Z(n)$$

Updating Step:

$$\hat{\mathbf{Y}}(n|n) = \hat{\mathbf{Y}}(n|n-1) + \mathbf{B}(n)(\mathbf{X}(n) - \hat{\mathbf{X}}(n|n-1))$$

$$\mathbf{B}(n) = \mathbf{R}(n|n-1)\mathbf{A}(n)^T[\mathbf{A}(n)\mathbf{R}(n|n-1)\mathbf{A}(n)^T + \mathbf{Q}_W(n)]^{-1}$$

$$\mathbf{R}(n|n) = (\mathbf{I} - \mathbf{B}(n)\mathbf{A}(n))\mathbf{R}(n|n-1)$$

Vector Kalman Filter with Control Input

Vector KF with Control Input

- In some applications, the time-evolution of process $\mathbf{Y}(n)$ can be affected by some known input, whose effect on $\mathbf{Y}(n)$ is known. This modifies the system model to:

$$\mathbf{Y}(n) = \mathbf{H}(n)\mathbf{Y}(n-1) + \mathbf{C}(n)\mathbf{u}(n) + \mathbf{Z}(n)$$

where

- $\mathbf{C}(n) \in \mathbb{R}^{r \times p}$ is the control-input model applied to $\mathbf{u}(n)$, assumed known.
- $\mathbf{u}(n) \in \mathbb{R}^p$ is the (known) control vector, representing a controlling input applied through the control-input model.
- The Kalman filter for such a system model is exactly the same as earlier, except the known control input is taken into account in the one-step prediction of $\mathbf{Y}(n)$ (see next slide)

Vector KF with Control Input

Correction for controlling input
in the prediction step



Prediction Step:

$$\hat{\mathbf{Y}}(n|n-1) = \mathbf{H}(n)\hat{\mathbf{Y}}(n-1|n-1) + \mathbf{C}(n)\mathbf{u}(n),$$

$$\hat{\mathbf{X}}(n|n-1) = \mathbf{A}(n)\hat{\mathbf{Y}}(n|n-1).$$

In addition, the covariance matrix of the estimation error of $\hat{\mathbf{Y}}(n|n-1)$ is:

$$\mathbf{R}(n|n-1) = \mathbf{H}(n)\mathbf{R}(n-1|n-1)\mathbf{H}(n)^T + \mathbf{Q}_Z(n)$$

Updating Step:

$$\hat{\mathbf{Y}}(n|n) = \hat{\mathbf{Y}}(n|n-1) + \mathbf{B}(n)(\mathbf{X}(n) - \hat{\mathbf{X}}(n|n-1))$$

with the Kalman gain:

$$\mathbf{B}(n) = \mathbf{R}(n|n-1)\mathbf{A}(n)^T[\mathbf{A}(n)\mathbf{R}(n|n-1)\mathbf{A}(n)^T + \mathbf{Q}_W(n)]^{-1}$$

In addition, the covariance matrix of the estimation error of $\hat{\mathbf{Y}}(n|n)$ is:

$$\mathbf{R}(n|n) = (\mathbf{I} - \mathbf{B}(n)\mathbf{A}(n))\mathbf{R}(n|n-1)$$

Initialization: the vector Kalman filter is initialized by setting $\hat{\mathbf{Y}}(0|0) = \boldsymbol{\mu}_{\mathbf{Y}(0)}$ and $\mathbf{R}(0|0) = \boldsymbol{\Sigma}_{\mathbf{Y}(0)\mathbf{Y}(0)}$